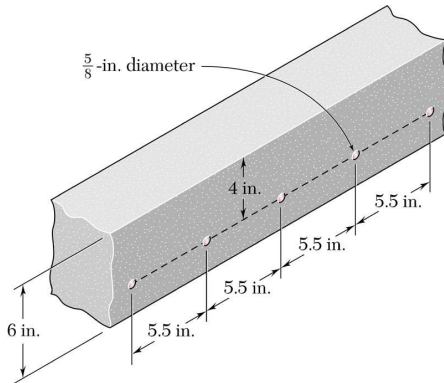


Zeshui Song, 3/28/26, HW6

Problem 4.51 will be graded.

4.51 A concrete slab is reinforced by 5/8-in.-diameter rods placed on 5.5-in. centers as shown. The modulus of elasticity is 3×10^6 psi for the concrete and 29×10^6 psi for the steel. Using allowable stresses of 1400 psi for the concrete and 20 ksi for the steel, determine the largest bending moment per foot of width that can be safely applied to the slab.



Given:

- $E_c = 3 \times 10^6$ psi
- $E_s = 29 \times 10^6$ psi
- $\sigma_{all,c} = 1400$ psi
- $\sigma_{all,s} = 20 \times 10^3$ psi
- $D = \frac{5}{8}$ in

Find:

- Max bending moment per foot of width for the slab

1

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Approach:

- I will look at just a single 5.5-inch section to find the moment per inch, then scale it up to moment per 12 in (1 foot).
- Assuming that the concrete only bears compressive load, we will only consider the concrete above the neutral axis
- Use a transformed section for the steel with concrete as the reference material

$$n = \frac{E_s}{E_c}, \quad A_s^{Tr} = nA_s$$

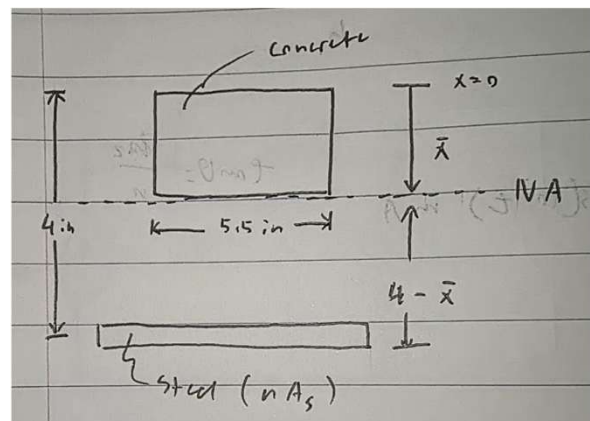
- I can locate the neutral axis relative to the arbitrary datum using the equation

$$\bar{x} = \frac{\sum A_i \bar{x}_i}{\sum A_i}$$

- Relate the max allowable stress to the critical moments in each material. Since concrete is the reference material, it doesn't need to be rescaled by n .

$$\sigma_c = -\frac{My}{I_{Tr}} \rightarrow \sigma_{max,c} = \sigma_{all,c} = -\frac{M_{max,c} \bar{x}}{I_{Tr}} \quad (\text{compressive})$$

$$\sigma_s = n \frac{My}{I_{Tr}} \rightarrow \sigma_{max,s} = \sigma_{all,s} = n \frac{M_{max,s} (4 - \bar{x})}{I_{Tr}} \quad (\text{tensile})$$



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Find transformed area (each 5.5-in section only contains 1 diameter of steel rod)

$$n = \frac{E_s}{E_c} = \frac{[29 \times 10^6 \text{ psi}]}{[3 \times 10^6 \text{ psi}]} = 9.6666$$

$$A_s^{Tr} = nA_s = n\left(\frac{\pi}{4}D^2\right) = 9.6666\left(\frac{\pi}{4}[0.625 \text{ in}]^2\right) = 2.9656 \text{ in}^2$$

Find the neutral axis

$$\bar{x} = \frac{\sum A_i \bar{x}_i}{\sum A_i} = \frac{([5.5 \text{ in}] \bar{x})\left(\frac{\bar{x}}{2}\right) + (nA_s)([4 \text{ in}])}{[5.5 \text{ in}] \bar{x} + nA_s} \rightarrow \bar{x} = \frac{2.75\bar{x}^2 + 11.8627}{5.5\bar{x} + 2.9656}$$

Solve the quadratic for $\bar{x}$

$$5.5\bar{x}^2 + 2.9656\bar{x} = 2.75\bar{x}^2 + 11.8627$$

$$\bar{x} = \frac{-2.9656 \pm \sqrt{(2.9656)^2 - 4(2.75)(-11.8627)}}{5.5} = 1.6065 \text{ in}, -2.6850 \text{ in}$$

Since $\bar{x}$ must be a positive value (neutral axis must be within the beam material): $\bar{x} = 1.6065 \text{ in}$

3

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Find the moment of inertia

$$I_{c,NA}^{Tr} = \frac{1}{12}bh^3 + A_c\left(\frac{\bar{x}}{2}\right)^2 = \frac{1}{12}[5.5 \text{ in}]\bar{x}^3 + [5.5 \text{ in}]\bar{x}\left(\frac{\bar{x}}{2}\right)^2 = 7.6012 \text{ in}^4$$

$$I_{s,NA}^{Tr} = \frac{1}{64}\pi D^4 + nA_s([4 \text{ in}] - \bar{x})^2 = \frac{1}{64}\pi[0.625 \text{ in}]^4 + [2.9656 \text{ in}^2]([4 \text{ in}] - \bar{x})^2 = 16.9969 \text{ in}^4$$

$$I_{NA}^{Tr} = I_{c,NA}^{Tr} + I_{s,NA}^{Tr} = 24.5981 \text{ in}^4$$

Find the critical moments

$$\sigma_{all,c} = -\frac{M_{max,c}\bar{x}}{I^{Tr}} \rightarrow M_{max,c} = \frac{\sigma_{all,c}I^{Tr}}{\bar{x}} = \frac{[1400 \text{ psi}][24.5981 \text{ in}^4]}{[1.6065 \text{ in}]} = 21436.2527 \text{ lbf} \cdot \text{in}$$

$$\sigma_{all,s} = n\frac{M_{max,s}(4 - \bar{x})}{I^{Tr}} \rightarrow M_{max,s} = \frac{\sigma_{all,s}I^{Tr}}{n(4 - \bar{x})} = \frac{[20 \times 10^3 \text{ psi}][24.5981 \text{ in}^4]}{9.6666[4 - 1.6065 \text{ in}]} = 21264.312 \text{ lbf} \cdot \text{in}$$

Pick the smaller critical moment:

$$M_{max} = 21264.312 \text{ lbf} \cdot \text{in}$$

Convert to moment per foot:

$$= \frac{[21264.312 \text{ lbf} \cdot \text{in}]}{[5.5 \text{ in}]} \cdot \frac{[12 \text{ in}]}{[1 \text{ ft}]} = 46394.8625 \frac{\text{lbf} \cdot \text{in}}{\text{ft}} (\text{moment per foot})$$

$$M_{max} = 46.394 \frac{\text{kip} \cdot \text{in}}{\text{ft}} (\text{moment per foot})$$

4

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Since the neutral axis is only 1.6065 in from the top, most of the concrete is not loaded under pure bending. Additionally, the max allowable moment for steel turned out to be less than the allowable moment for concrete, therefore it is likely that the steel will start to yield/fail before the concrete does. [Time spent: 1hr]

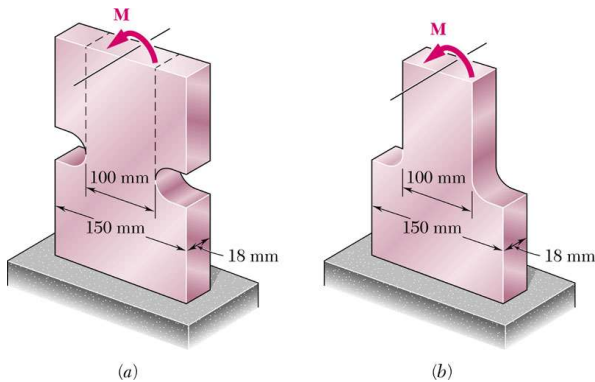
5

Zeshui Song, 3/28/26, HW6

4.65 The allowable stress used in the design of a steel bar is 80 MPa. Determine the largest couple **M** that can be applied to the bar (a) if the bar is designed with grooves having semicircular portions of radius $r = 15\text{ mm}$, as shown in figure a, (b) if the bar is redesigned by removing the material above the grooves as shown in figure b.

- Given:
- $\sigma_{all} = 80\text{ MPa}$
 - $r = 15\text{ mm}$

- Find:
- Max moment couple **M** on the bar for (a) and (b)



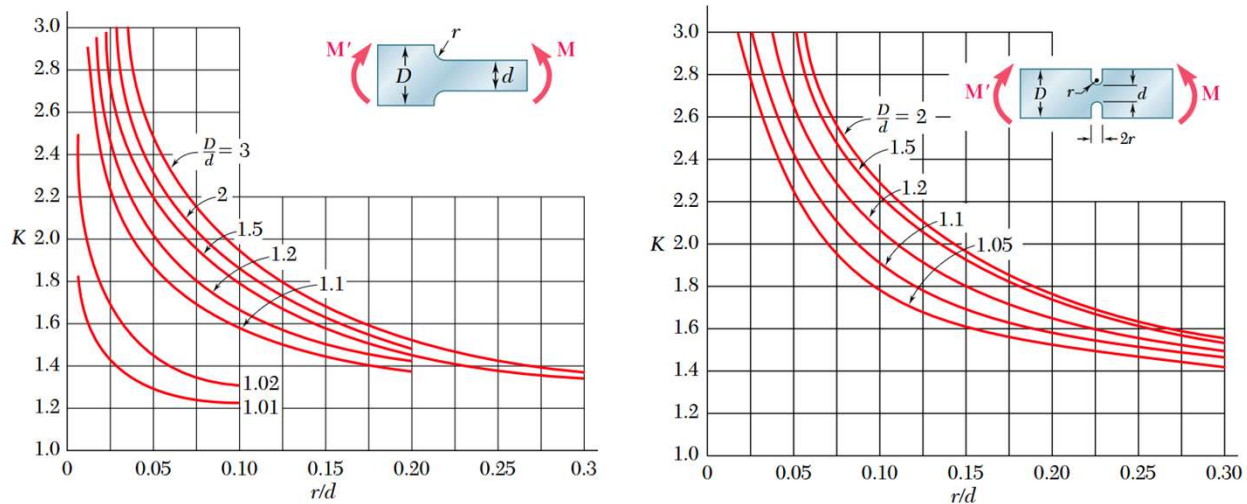
- Approach:
- The critical section of 100 mm width is the same for both cases, so the only thing that changes is the stress concentration factor (found on the plots)
 - Using the stress concentration factor, I can relate the max allowable stress to the max moment couple

$$\sigma_{max} = \sigma_{all} = K \frac{M_{max}c}{I}$$

Both c and I refer to the cross section at the critical section.

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7

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Find the stress concentration factors

$$\frac{D}{d} = \frac{[150 \text{ mm}]}{[100 \text{ mm}]} = 1.5 \quad K_a = 1.93$$

$$\frac{r}{d} = \frac{[15 \text{ mm}]}{[100 \text{ mm}]} = 0.15 \quad K_b = 1.57$$

Moment of inertia

$$I = \frac{1}{12}bh^3 = \frac{1}{12}[18 \text{ mm}][100 \text{ mm}]^3 = 1.5 \times 10^6 \text{ mm}^4 = 1.5 \times 10^{-6} \text{ m}^4$$

Find M

$$\sigma_{all} = K \frac{M_{max}c}{I} \rightarrow M_{max} = \frac{\sigma_{all}I}{Kc}$$

$$(a) \quad M_{max} = \frac{\sigma_{all}I}{Kc} = \frac{[80 \times 10^6 \text{ Pa}][1.5 \times 10^{-6} \text{ m}^4]}{[1.93][50 \times 10^{-3} \text{ m}]} = 1243.523 \text{ N} \cdot \text{m}$$

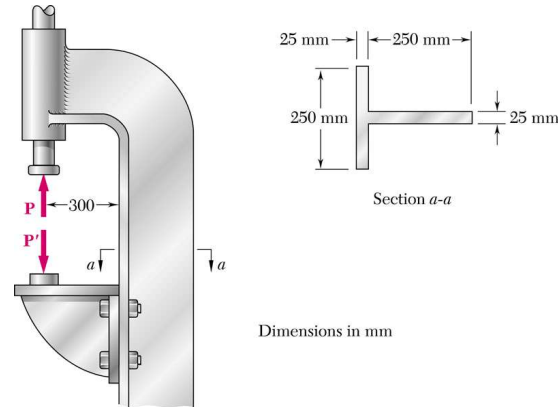
$$(b) \quad M_{max} = \frac{\sigma_{all}I}{Kc} = \frac{[80 \times 10^6 \text{ Pa}][1.5 \times 10^{-6} \text{ m}^4]}{[1.57][50 \times 10^{-3} \text{ m}]} = 1528.662 \text{ N} \cdot \text{m}$$

The stress concentration factor is lower for b which makes sense why the max moment couple is higher for b. Since the material and geometry of the critical region is the same, the stress concentrator is the only thing that can affect how the max moment couples compare. [Time spent: 20 mins]

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Zeshui Song, 3/28/26, HW6

4.113 Knowing that the allowable stress in section a-a of the hydraulic press shown is 40 MPa in tension and 80 MPa in compression, determine the largest force P that can be exerted by the press.



Given:

- $\sigma_{all,t} = 40 \text{ MPa}$
- $\sigma_{all,c} = 80 \text{ MPa}$

Find:

- Largest force P that can be exerted

Approach:

- Using superposition, I can determine the normal stress due to bending and due to axial loading separately and then sum them up.
- First using a FBD, I can find the internal reaction moment due to P acting on the beam. The max stresses caused by this moment is

$$\sigma_{m,c} = -\frac{Mc}{I}, \sigma_{m,t} = \frac{M(h-c)}{I}$$

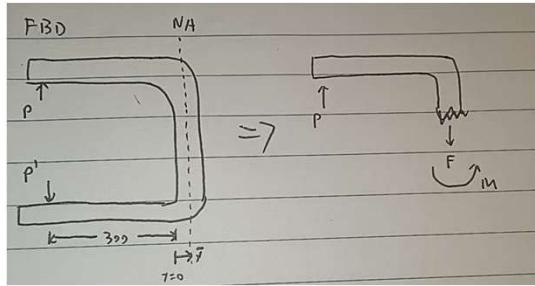
- Using the FBD I can find the internal reaction forces due to the load P . The stress caused by the axial load is then

$$\sigma_P = \frac{F}{A}$$

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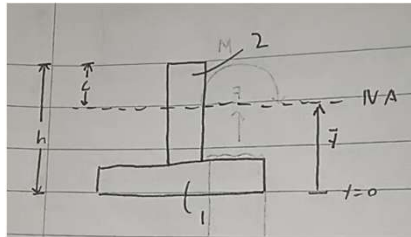
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FBD (assuming massless)



$$\sum F = P - F = 0 \rightarrow P = F$$

$$\sum M_{NA} = -P([300 \text{ mm}] + \bar{y}) + M \rightarrow M = P([300 \text{ mm}] + \bar{y})$$



Find the neutral axis

$$\bar{y} = \frac{\sum A_i \bar{y}_i}{\sum A_i} = \frac{[6250 \text{ mm}^2][12.5 \text{ mm}] + [6250 \text{ mm}^2][150 \text{ mm}]}{[6250 + 6250 \text{ mm}^2]} = 81.25 \text{ mm}$$

Thus the moment is

$$M = P[300 + 81.25 \text{ mm}] = P[0.38125 \text{ m}]$$

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Find the moment of inertia using superposition

$$I_{1,NA} = \frac{1}{12} b_1 h_1^3 + A_1 (\bar{y} - \bar{y}_1)^2 = \frac{1}{12} [250 \text{ mm}] [25 \text{ mm}]^3 + [6250 \text{ mm}^2] [81.25 - 12.5 \text{ mm}]^2 = 29.8665 \times 10^6 \text{ mm}^4$$

$$I_{1,NA} = 29.8665 \times 10^{-6} \text{ m}^4$$

$$I_{2,NA} = \frac{1}{12} b_2 h_2^3 + A_2 (\bar{y}_2 - \bar{y})^2 = \frac{1}{12} [25 \text{ mm}] [250 \text{ mm}]^3 + [6250 \text{ mm}^2] [150 - 81.25 \text{ mm}]^2 = 62.093 \times 10^6 \text{ mm}^4$$

$$I_{2,NA} = 62.093 \times 10^{-6} \text{ m}^4$$

$$I = I_{1,NA} + I_{2,NA} = 91.9595 \times 10^{-6} \text{ m}^4$$

Geometry

$$h = 250 + 25 \text{ mm} = 275 \text{ mm}$$

$$c = h - \bar{y} = 275 - 81.25 \text{ mm} = 193.75 \text{ mm}$$

Solve for the area

$$A = 2 \cdot 6250 \text{ mm}^2 = 12500 \text{ mm}^2 = 0.0125 \text{ m}^2$$

Formula for total stress at the a-a cross section:

$$\sigma_{max,c} = -\frac{Mc}{I} + \frac{F}{A} = -\frac{P[0.38125 \text{ m}]c}{I} + \frac{P}{A} \text{ (compressive)}$$

$$\sigma_{max,t} = \frac{M(h-c)}{I} + \frac{F}{A} = \frac{P[0.38125 \text{ m}](h-c)}{I} + \frac{P}{A} \text{ (tensile)}$$

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Solving max P for compressive case:

$$\sigma_{max,c} = \sigma_{all,c} = -\frac{P[0.38125 \text{ m}]c}{I} + \frac{P}{A} = -\frac{P[0.38125 \text{ m}][0.19375 \text{ m}]}{[91.9595 \times 10^{-6} \text{ m}^4]} + \frac{P}{[0.0125 \text{ m}^2]} = -723.2578P$$

$$\sigma_{all,c} = [80 \times 10^6 \text{ Pa}] = 723.2578P$$

$$P = 110610.625 \text{ N}$$

Solving max P for tensile case:

$$\sigma_{max,t} = \sigma_{all,t} = \frac{P[0.38125 \text{ m}](h-c)}{I} + \frac{P}{A} = \frac{P[0.38125 \text{ m}][0.08125 \text{ m}]}{[91.9595 \times 10^{-6} \text{ m}^4]} + \frac{P}{[0.0125 \text{ m}^2]} = 416.85P$$

$$\sigma_{all,t} = [40 \times 10^6 \text{ Pa}] = 416.85P$$

$$P = 95957.76 \text{ N}$$

Picking the smaller P:

$$P = 95.957 \text{ kN}$$

This makes sense since the neutral axis is closer to the compressive side, so the max compressive stress would've been small, especially after superposing the tensile axial stress to it. Thus, the max tensile stress ends up being higher because the effects are additive. Since the max allowable tensile stress is also much smaller (40 MPa), it makes sense that it is the limiting case. [Time spent: 45 mins]

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